

Li-Fi Data Transmitter Using Visible Light Communication

Transmission of Digital Data through LED Light using Arduino Microcontrollers

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Project Category

Electronics • Embedded Systems • Optical Communication

Abstract

Li-Fi (Light Fidelity) is an emerging wireless communication technology that uses visible light instead of radio waves for transmitting digital information. This project demonstrates a basic Li-Fi communication system capable of transmitting keypad inputs from one Arduino-based transmitter to another Arduino-based receiver using an LED as the communication medium.

The transmitter consists of an Arduino Uno connected to a 4×4 keypad and an LED. Whenever a key is pressed, the Arduino converts the key into a unique pulse duration and flashes the LED accordingly. At the receiving end, an LDR detects the variations in light intensity. An Arduino Nano measures the received pulse duration, decodes the transmitted character, and displays the received output on a 16×2 LCD display.

The project illustrates the concept of Visible Light Communication (VLC), where information is encoded in rapid changes in LED brightness that are too fast for the human eye to perceive. The system was designed, assembled, tested, and optimized through multiple iterations to improve transmission accuracy and reliability.

Introduction — What is Li-Fi?

Li-Fi (Light Fidelity) is a wireless communication technology that uses Visible Light Communication (VLC) to transmit digital information. Unlike Wi-Fi, which transfers data through radio frequency (RF) waves, Li-Fi uses rapid modulation of LED light intensity.

The blinking of an LED occurs at extremely high speed, making it invisible to the human eye while allowing electronic sensors such as photodiodes or Light Dependent Resistors (LDRs) to detect the encoded information.

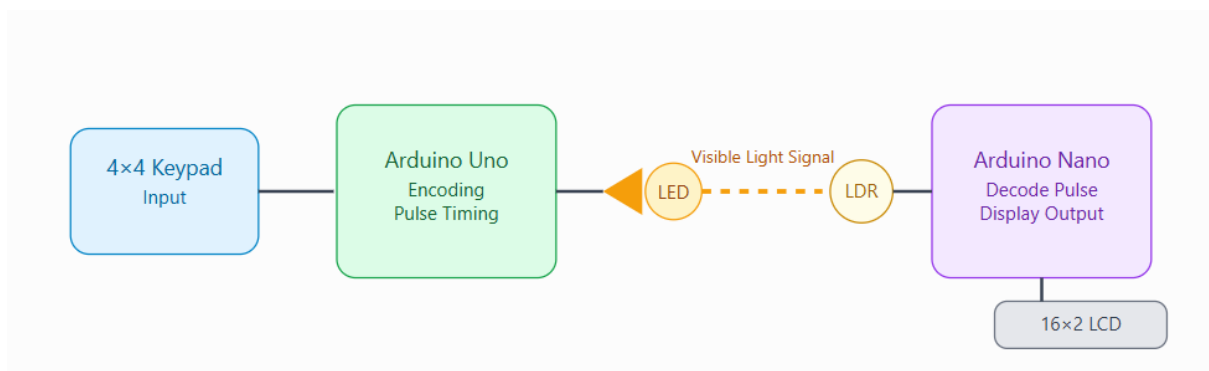
Objectives

- Design a wireless communication system using visible light.
- Transmit keypad inputs through LED light pulses.
- Detect optical signals using an LDR sensor.
- Decode received signals using Arduino Nano.
- Display received data on a 16×2 LCD module.
- Demonstrate the working principle of Visible Light Communication.

Components Required

Component	Purpose
Arduino Uno	Transmitter controller.
Arduino Nano	Receiver controller.
4×4 Matrix Keypad	Input device.
High Brightness LED	Light transmitter.
LDR (Light Dependent Resistor)	Light receiver.
16×2 LCD with I2C Module	Display received character.
220Ω Resistor	LED current limiting.
Breadboard	Circuit assembly.
Jumper Wires	Connections.
USB Power Supply	Powering Arduinos.

System Architecture



Operational Flow

Step 1 — Input at Transmitter

Step 2 — Data Encoding

Step 3 — Optical Transmission

Step 4 — Signal Detection

Step 5 — Decoding & Display



Future Scope

- High-speed internet through ceiling LEDs.
- Vehicle-to-vehicle communication.
- Smart street lighting.
- Hybrid Wi-Fi + Li-Fi networks.
- Indoor positioning systems.
- Radio sensitive areas in hospitals

Conclusion

The Li-Fi Data Transmitter project successfully demonstrates wireless data transmission using visible light instead of radio waves. By integrating Arduino Uno, Arduino Nano, LEDs, LDR sensors, keypad input, and LCD output, the project proves the practical implementation of Visible Light Communication on a small scale.

The system reliably transmits digital keypad inputs through optical pulses and decodes them at the receiving end. This project provides a strong foundation for understanding optical wireless communication and embedded system design.